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Harmful effect of climate change on the biodiversity of some avian species of Kanpur city

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Abstract : Change of environment is a very serious problem for the survival of living organisms specially birds that affects many physiological activities of their body. There are many biotic and abiotic factors which are responsible for decline of the bird's habitat in the environment. The temperature, rainfall and availability of the food are the main important factors which affects biodiversity of avian species in the nature. Climate change is the result of many human activities like deforestation, excessive use of energy and over use of fossil fuels. There are some other natural disturbances like volcanic eruptions, solar variability, etc. which directly affects the ozone layer present in the atmosphere. It is another source of climatic change in the nature and is totally responsible for the decline of avian fauna in the natural surroundings.

Present study focuses on harmful effects of climatic changes on the biodiversity of some avian species of Kanpur. The data of some native, domestic and wild species of birds were collected in the month of July and August, November and December and April and May 2011. It is clearly evident from the above study that the variety of birds is declining and is becoming extinct day by day due to climate changes in the environment by many anthropogenic activities.

Key Words: Volcanic eruptions, solar variability, avian fauna, anthropogenic activities.

INTRODUCTION

Environmental change is a very complex phenomenon that affects many physiological activities in many living organisms directly or indirectly specially in birds. There are so many physical, chemical and biological factors which are responsible for decline of bird's habitat in the natural environment (Root, 1988a; 1988b)[1,2]. The sudden change in the environment is one of the major problem, which has affected the population dynamics of birds in the nature (Crick *et al.* 1993; Newton, 1998)[3,4]. There are many scientific reasons and evidences available, which support that birds have been migrating from one place to another due to climate change in response to global warming for their survival (Permesan and Yohe, 2003; Lack, 1954 and 1966)[5,6,7]. The impact of climate change on the biodiversity of *Avian fauna* has been a major field of research area by the ornithologists all over the world (Crick, 1999; Root, 1988a; Root, 1988b)[8,1,2]. Change of climate disturbs the ecological responses, which is very important phenomenon and essential factor for the survival of birds in natural conditions. The change of climate disturbs physiological response in the internal state of the body. The change of temperature is very important abiotic factor which is very essential for the survival of birds in the natural habitat (Walther *et al.,* 2002)[9].

According to Walther *et al.* (2002)[9], the change of environment disturbs the natural cycle in variety of birds. There are so many physiological changes which has occurred in the body and which caused distribution of birds in the environment (Gibbons *et al.* 1993)[10].

The natural selection caused due to sudden change of environment (Lack 1954; 1966)[6,7]. There are many workers explained very clearly time to time and explained the importance of natural habitat in the dense forest region.

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According to them climate change effects the metabolic rate of birds. The behavioural change in variety of birds like feeding, courtship and mating behaviour influence indirectly or directly due to environmental changes created by man by the many anthropogenic activities (Crick *et al.,* 1993; Gibbons *et al.,* 1993; Stenseth *et al.,* 2002 and Berry *et al.,* 2001).

There are so many abiotic and biotic factors which are responsible to change and disturb the ecological balance in biological communities specially the native and wild species of birds in the nature. Thus, the change of environment badly affected and caused to decline the migratory and native species of birds (Wrinkler *et al.,* 2002).

The climate change in environment directly affects so many behavioural patterns in birds, like earlier breeding, egg size, change in timing of migration and change of distribution patterns. Due to the change of environment, the biodiversity of *Avian fauna* restricted in specific areas (Lack, 1966; Newton, 1998).

Thus climatic change directly effects the distribution and geographical range in variety of domestic and wild birds (Root *et al.,* 2003). It is observed that when a particular species of birds is accurately mapped it require possible to correlate the presence or absence of species in a specific areas due to environmental variations.

The main objective of the present study is to observe, the harmful effects of environmental change, on the biodiversity of terrestrial *Avian fauna* of Kanpur City. Due to sudden change of climate, the ecological cycle of variety of birds got badly effected. In present situation environmental changes are responsible to decline the native, wild and migratory species of birds in the nature (Gibbons *and Chapman, 1993*; Thomas and Lennon, 1999).

MATERIAL AND METHODS

For the study of harmful effect of climatic change in the environment, on the biodiversity of some *avian fauna* in Kanpur City, a general survey was conducted from the three sites in the Kanpur City. These sites were Nawabganj (Zoo Area), Kalyanpur-Mainawati Marg and Industrial-Tannery Area of Jajmau. The survey was conducted in the Rainy season i.e. (July and August, 2011), winter (November 2011 and December, 2011), and Summer season i.e. (April & May, 2011).

The data were collected from the three sites in the city were wild, domestic and agricultural farmland *avian* species of Kanpur City during winter, summer and rainy season. Just after the data collection, the name and the species of birds were arranged systematically in tabular form to compare and find out that which species of bird declines or at the state of extinction day by day due to sudden change of climate in the natural condition of environment.

RESULTS AND DISCUSSION

The result of present study shows impact of harmful effects of climatic changes in the biodiversity of *avian fauna* in the Kanpur City. For this purpose, survey of three sites was taken. These sites were Nawabganj (Zoo Area), Kalyanpur-Mainawati Marg and Industrial Tannery Area of Jajmau. It is observed that most bird species found in Nawabganj Zoo-Area (22) and Kalyanpur-Mainawati Marg (24) in the winter, summer and rainy season of the year (Table I and II). It is also concluded that in the Industrial Tannery Area of Jajmau minimum bird species (06) were recorded (Table III).

There are some endangered species of birds like Quails, Vulture Crain and Owlet, which were never seen at the time of survey area during the winter, summer and rainy season of the year. These bird species were decline day by day or at the stage of extinction due to climatic changes in the environment.

The result of the present study clearly concluded, that change of environment due to many anthropogenic activities are main factor for decline, the biodiversity of *avian fauna* in the city (Walther *et al.,* 2002; Thomas and Lennon, 1999). It was also observed that data which was collected in Nawabganj (Zoo Area) and Kalyanpur-Mainawati Marg. There are (22) and (24) bird species were seen due to presence of green vegetation, agricultural farmland areas which are suitable habitat for the birds. The green plants, trees are good climatic condition for the survival of birds.

In the Industrial Tannery Area of Jajmau, environmental condition are not suitable for the survival of the birds, there were no green vegetation and there were deforestation for the industrial purpose and smoke of factories pollute the environment all around the city. In this area only (06) Bird species were recorded there.

Therefore, data showed the presence of harmful effect in the biodiversity of *avian* species.

According to (Stenseth *et. al.,* 2002; Wrinkler *et al.,* 2002 and Root and Schheider, 1993), the ecology of *avian fauna* was affected due to climate fluctuations in the nature. The nesting, nutritive and breeding behaviour of birds were influenced by environmental changes. The reproductive cycle of birds were also get influenced by change of climate.

Finally, from the results of present work it can be concluded that, there is a great impact of climatic change on the biodiversity of *avian fauna* of Kanpur region. The sudden change of environment due to deforestation, over use of fuels and other anthropogenic activities by human beings are responsible for the disturbances of *avian* species in the nature (Perminesan and Yohe, 2003; Wrinkler *et al.,* 2002).

Table – I: Showing the bird species in Nawabganj (Zoo Area) of Kanpur City in the rainy, winter and summer season.

S.No.	Name of Bird	July & Aug., 2011	Nov. & Dec., 2011	April & May, 2011
1.	Pavo cristatus (Peacock or Mor)	✓	×	✓
2.	Columba livia (Pigeon)	✓	✓	✓
3.	Eudynamys (Koel)	✓	✓	✓
4.	Psittacule (Parrot)	✓	✓	✓
5.	Coracias (Nilkanth)	×	×	✓
6.	Passer domesticus (House Sparrow)	✓	×	✓
7.	Corvus splendens (Crow)	✓	✓	✓
8.	Geese	×	×	×
9.	Vulture	×	×	×
10.	Dinopium (Wood pecker)	×	×	✓
11.	Quails	×	×	×
12.	Milvus-migraus (Kite)	×	×	×
13.	Gypus (Gidh)	×	×	×
14.	Phoenicopterus (Flamingo)	✓	×	×
15.	Bubo-bubo (Owl or Ghughu)	×	×	✓
16.	Swan	×	×	×
17.	Duck	✓	×	✓

Where : × = Bird species not seen in the survey area. **Total = 22**

✓ = Bird species seen in the survey area.

Table – II: Showing the bird species in Kalyanpur-Mainawati Marg of Kanpur City in the rainy, winter and summer season

S. No.	Name of Bird	July & Aug., 2011	Nov. & Dec., 2011	April & May, 2011
1.	Pavo cristatus (Peacock or Mor)	✓	×	✓
2.	Columba livia (Pigeon)	✓	✓	✓
3.	Eudynamys (Koel)	✓	✓	✓
4.	Psittacule (Parrot)	✓	✓	✓
5.	Coracias (Nilkanth)	×	×	×
6.	Passer domesticus (House Sparrow)	✓	×	✓
7.	Corvus splendens (Crow)	✓	✓	✓
8.	Geese	×	×	×
9.	Vulture	×	×	×
10.	Dinopium (Wood pecker)	✓	×	✓
11.	Quails	×	×	×
12.	Milvus-migraus (Kite)	×	×	✓
13.	Gypus (Gidh)	×	×	×
14.	Phoenicopterus (Flamingo)	✓	×	×
15.	Bubo-bubo (Owl or Ghughu)	×	×	×
16.	Swan	×	✓	×
17.	Duck	✓	✓	✓

Where : × = Bird species not seen in the survey area. **Total = 24**
✓ = Bird species seen in the survey area.

Table – III: Showing the bird species in Industrial Tannery Area of Jajmau of Kanpur City in the rainy, winter & summer season.

S.No.	Name of Bird	July & Aug., 2011	Nov. & Dec., 2011	April & May, 2011
1.	Pavo cristatus (Peacock or Mor)	×	×	×
2.	Columba livia (Pigeon)	×	×	×
3.	Eudynamys (Koel)	×	×	×
4.	Psittacule (Parrot)	✓	×	×
5.	Coracias (Nilkanth)	×	×	×
6.	Passer domesticus (House Sparrow)	✓	×	✓
7.	Corvus splendens (Crow)	✓	✓	✓
8.	Geese	×	×	×
9.	Vulture	×	×	×
10.	Dinopium (Wood pecker)	×	×	×
11.	Quails	×	×	×
12.	Milvus-migraus (Kite)	×	×	×
13.	Gypus (Gidh)	×	×	×
14.	Phoenicopterus (Flamingo)	×	×	×
15.	Bubo-bubo (Owl or Ghughu)	×	×	×
16.	Swan	×	×	×
17.	Duck	×	×	×

Where : × = Bird species not seen in the survey area. **Total =06**
✓ = Bird species seen in the survey area.

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